

Discrete Mathematical Structures (BCS405A)

Question Bank

Module 3: Functions

- Define Cartesian product of two sets. Let $A = \{1, 2, 3\}$, $B = \{2, 3\}$ & $C = \{1, 3, 4\}$. Find the Following
 (i) $A \times (B \cup C)$ (ii) $(A \times B) \cup (A \times C)$ (iii) $A \times (B \cap C)$ (iv) $(A \times B) \cap (A \times C)$ (v) $A \times (B - C)$
 (vi) $(A \times B) - (A \times C)$ (vii) $|A \times B|$ (viii) $|A \times A|$.
- For non-empty sets A, B, C prove that, (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$
 (ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$ (iii) $A \times (B - C) = (A \times B) - (A \times C)$
- a) Define Function. Determine which of the following relations defined from $A = \{1, 2, 3, 4\}$ to $B = \{a, b, c\}$ are functions
 i) $R_1 = \{(1, a), (2, b), (3, c), (4, a)\}$ ii) $R_2 = \{(1, a), (2, a), (3, a), (4, a)\}$
 iii) $R_3 = \{(1, a), (2, b), (3, c)\}$ iv) $R_4 = \{(1, a), (1, b), (2, b), (3, c), (4, a)\}$
 b) Determine which of the following relations are functions. If a relation is a function, find its range.
 i) $R_1 = \{(x, y) \mid x, y \in Q, x^2 + y^2 = 1\}$, where Q is set of all rational numbers.
 ii) $R_2 = \{(x, y) \mid x^2 = y\} \subseteq \mathbb{R} \times \mathbb{R}$, where $\mathbb{R}$ is set of all reals. iii) $R_3 = \{(x, y) \mid x = y^2\} \subseteq \mathbb{R} \times \mathbb{R}$
 iv) $R_4 = \{(x, y) \mid y = 3x + 1\} \subseteq \mathbb{Z} \times \mathbb{Z}$ v) $R_5 = \{(x, y) \mid y = x^2 + 1\} \subseteq \mathbb{Z} \times \mathbb{Z}$
 c) Let $A = \{1, 2, 3, 4\}$ and $f : A \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 - x + 1$. Find the range of f .
- a) Define image set. Let $A = \{1, 2, 3, 4, 5\}$ & $B = \{w, x, y, z\}$. Find the images of the following under f , if $f : A \rightarrow B$ is $f = \{(1, w), (2, x), (3, x), (4, y), (5, y)\}$:
 $A_1 = \{1\}$, $A_2 = \{2, 3\}$, $A_3 = \{3, 4\}$, $A_4 = \{2, 3, 4, 5\}$
 b) Let a function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$. Find the images of $A_1 = \{2, 3\}$, $A_2 = \{-2, 0, 3\}$, $A_3 = (0, 1)$, $A_4 = (-3, 3)$, $A_5 = [-3, 2)$ and $A_6 = (-4, -3] \cup [5, 6]$.
- a) Define preimage set. Let $A = \{1, 2, 3, 4, 5, 6, 7\}$, $B = \{2, 4, 6, 8, 10, 12\}$ and $f : A \rightarrow B$ is given by $f = \{(1, 2), (2, 6), (3, 6), (4, 8), (5, 6), (6, 8), (7, 12)\}$. Determine preimages of the following subsets of B under f , $B_1 = \{4, 10\}$, $B_2 = \{2\}$, $B_3 = \{6\}$, $B_4 = \{6, 8, 12\}$, $B_5 = \{2, 6, 8, 12\}$.
 b) Let a function $f : \mathbb{Z} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 + 5$. Determine $f^{-1}(B)$ for the following $B \subseteq \mathbb{R}$.
 i) $B = \{6\}$, ii) $B = \{2, 6\}$, iii) $B = [6, 7]$, iv) $B = [6, 10]$, v) $B = [-4, 5]$ vi) $B = [-4, 5]$ and vii) $B = [5, \infty)$.
 c) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 3x - 5 & \text{for } x > 0 \\ -3x + 1 & \text{for } x \leq 0 \end{cases}$. Determine i) $f(0)$, $f(-1)$, $f\left(\frac{5}{3}\right)$, $f\left(-\frac{5}{3}\right)$
 ii) $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(-1)$, $f^{-1}(3)$, $f^{-1}(-3)$, $f^{-1}(-6)$, $f^{-1}(\{1, 2\})$, $f^{-1}([-5, 5])$, $f^{-1}([-6, 5])$.
- a) How many functions are possible from A to B in each of the following cases
 i) $|A| = m$ & $|B| = n$ ii) $A = \{1, 2, 3, 4\}$ & $B = \{a, b, c\}$ iii) $A = \{1, 2, 3, 4\}$ & $B = \{a, b, c, d, e\}$
 iv) $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$ and image of 1 is a .
 b) If there are 2,187 functions from A to B and $|B| = 3$, then determine $|A|$.
 b) Let $A = \{1, 2, 3, 4, 5\}$, $B = \{a, b, c, d\}$, $A_1 = \{2, 3, 4\} \subseteq A$ and $g : A_1 \rightarrow B$. In how many ways can g be extended to function $f : A \rightarrow B$.

7. Define one-to-one function and on to function. Determine whether the following functions are bijective, also find their ranges:

a) Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$ & $C = \{w, x, y, z\}$.

i) $f = \{(1, w), (2, x), (3, y), (4, z)\}$ ii) $f = \{(1, a), (2, b), (3, c), (4, c)\}$ iii) $f = \{(a, w), (b, x), (3, y)\}$

iv) $f = \{(1, w), (2, w), (3, w), (4, w)\}$ v) $f = \{(1, w), (2, x), (3, y), (4, w)\}$

b) i) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 2x + 1$ ii) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 1$ iii) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(a) = a + 1$

iv) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ v) $f : \mathbb{R} \rightarrow [0, \infty), f(x) = x^2$ vi) $f : [0, \infty) \rightarrow [0, \infty), f(x) = x^2$

vii) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$ viii) $f : \mathbb{R} \rightarrow (0, \infty), f(x) = e^x$ ix) $f : [0, \pi] \rightarrow \mathbb{R}, f(x) = \sin x$

x) $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}, f(x) = \sin x$ xi) $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1], f(x) = \sin x$

xii) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^2 + x$ xiii) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^3$ xiv) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^3 - x$

8. a) Let $A = \{1, 2, 3, 4\}$ & $B = \{a, b, c\}$. How many one-to-one functions are possible

i) from set A to set B ? ii) from set B to set A ? iii) from set B to set A with image of a is 1?

iv) from set B to set A with image of a is 1 and image of b is 2?

b) If $|A| = m$ & $|B| = n$, then how many one-to-one functions are possible from A to B ?

c) If $|A| = m$ & $|B| = n$, then how many bijective functions are possible from A to B ?

9. a) If $|A| = m$ & $|B| = n$, then how many onto functions are possible from A to B ?

b) Let $A = \{1, 2, 3, 4\}$ & $B = \{a, b, c\}$. How many onto functions are possible i) from set A to set B ?

ii) from set B to set A ?

c) Define stirling number of second kind. Compute $S(5, 3)$, $S(7, 2)$ & $S(8, 5)$

d) Prove that for $1 < n \leq m$, $S(m+1, n) = S(m, n-1) + nS(m, n)$. Further, if

$S(8, 4) = 1701$, $S(8, 5) = 1050$

& $S(8, 6) = 266$, evaluate $S(10, 6)$.

e) Find the number of ways of distributing 6 objects among 4 identical containers with some containers possibly left empty.

f) Find the number of ways of distributing 4 distinct objects among 3 identical containers with some containers possibly left empty.

g) Suppose we have seven different colored balls and four containers numbered I, II, III and IV.

i) In how many ways can we distribute the balls so that no container left empty? ii) In this collection of seven colored balls, one of them is blue. In how many ways can we distribute the balls so that no container is empty and the blue ball is in container II? iii) If we remove the number from the containers, so that no longer distinguish them, in how many ways can we distribute the balls among four identical container(s) possibly empty?

10. Let $A = \{1, 2, 3, 4\}$ & $B = \{1, 2, 3, 4, 5, 6\}$. i) How many functions are possible from set A to set B ? How many of these are one-to-one? How many are onto?

ii) How many functions are possible from set B to set A ? How many of these are onto? How many are one-to-one?

11. Let $f : A \rightarrow B$ and $A_1, A_2 \subseteq A$, then prove the following:

i) If $A_1 \subseteq A_2$, then $f(A_1) \subseteq f(A_2)$ ii) $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$ iii) $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$

iv) $f(A_1 \cap A_2) = f(A_1) \cap f(A_2)$, if f is one-to-one.

12. If $f : A \rightarrow B$ and $B_1, B_2 \subseteq B$, then prove the following:

$$\text{i) } f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2) \quad \text{ii) } f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2) \quad \text{iii) } f^{-1}(\overline{B_1}) = \overline{f^{-1}(B_1)}$$

13. Let $f, g: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$, where for all $x \in \mathbb{Z}^+$, $f(x) = x+1$ & $g(x) = \max\{1, x-1\}$. i) What is the range of f & g ? ii) is f a onto function? iii) is f one-to-one? Iv) is g one-to-one function v) is g an onto function?

14. Define Identity function and equal functions with an example for each

15. Define composite function. Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, $C = \{w, x, y, z\}$ with $f: A \rightarrow B$ & $g: B \rightarrow C$ be defined by $f = \{(1, a), (2, b), (3, c), (4, c)\}$ and $g = \{(a, w), (b, x), (c, z)\}$. Compute $g \circ f$ & $f \circ g$.

16. a) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$ & $g(x) = x+5$. Determine $g \circ f$ & $f \circ g$. Show that the composition of two functions is not commutative.

b) If $f: A \rightarrow B$, $g: B \rightarrow C$ and $h: C \rightarrow D$ are three functions then prove that $h \circ (g \circ f) = (h \circ g) \circ f$.

c) Let $f, g, h: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$, $g(x) = x+5$ & $h(x) = \sqrt{x^2+2}$. Prove $(h \circ g) \circ f = h \circ (g \circ f)$

d) Let functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = ax+b$ and $g(x) = 1-x+x^2$. If $(g \circ f)(x) = 9x^2 - 9x + 3$, determine the constants a, b .

e) Let $A = \{1, 2, 3, 4\}$ and $f: A \rightarrow A$ be defined by $f = \{(1, 2), (2, 2), (3, 1), (4, 3)\}$. Find f^2 , f^3 , f^4 & f^5

g) Let f, g, h be functions from $\mathbb{Z}$ to $\mathbb{Z}$ defined by $f(x) = x-1$, $g(x) = 3x$, $h(x) = \begin{cases} 0 & \text{if } x \text{ is even} \\ 1 & \text{if } x \text{ is odd} \end{cases}$

Determine (i) f^2 , f^3 , f^{500} , g^2 , g^3 , g^{500} , h^2 , h^3 , h^{500} . (ii) $g \circ f$, $h \circ g$, $g \circ h$, $f \circ g$.

Show that $h \circ (g \circ f) = (h \circ g) \circ f$ & $f \circ (g \circ h) = (f \circ g) \circ h$.

17. Let $f: A \rightarrow B$ and $g: B \rightarrow C$. Prove the following

a) i) If f & g are one-to-one, then $g \circ f: A \rightarrow C$ is one-to-one ii) If f & g are onto, then $g \circ f: A \rightarrow C$ is onto.

b) i) If $g \circ f$ is one-to-one, then f are one-to-one ii) If $g \circ f: A \rightarrow C$ is onto, then g are onto.

18. a) Define invertible function. Show that inverse of a function is unique, if it exists.

b) Prove that a function $f: A \rightarrow B$ is invertible if and only if it is one-to-one and onto.

c) Prove that $f: A \rightarrow B$ and $g: B \rightarrow C$ are invertible function then $g \circ f: A \rightarrow C$ is an invertible function and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$

19. a) Show that the following functions are invertible and find their inverses. i) $A = \{x \in \mathbb{R} \mid x \geq -1\}$ & $B = \{x \in \mathbb{R} \mid x \geq 0\}$, define $f: A \rightarrow B$ by $f(x) = \sqrt{x+1}$ ii) $f: \mathbb{R} \rightarrow \mathbb{R}^+$ by $f(x) = e^x$

b) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x+5$ & $g(x) = (x-5)/2$. Show that f & g are invertible.

c) Let $A = B = \mathbb{R}$ be the set of all real numbers, the function $f: A \rightarrow B$ and $g: B \rightarrow A$ be defined by $f(x) = 2x^3 - 1$, $\forall x \in A$; $g(y) = \left\{\frac{1}{2}(y+1)\right\}^{\frac{1}{3}}$, $\forall y \in B$. Show that each of f and g is the inverse of each other

d) Let $A = B = C = \mathbb{R}$, $f: A \rightarrow B$, $f(a) = 2a+1$; $g: B \rightarrow C$, $g(b) = b/2$. Compute $g \circ f$ and show that it is invertible.

20. State pigeonhole principle.

(i) An office employs 13 clerks. Show that at least 2 of them will have birthdays during the same month.

(ii) Let $\triangle ABC$ be an equilateral triangle of side 1. Show that if we select 10 points in the interior, there must be at least two points whose distance apart is less than $\frac{1}{3}$.

- (iii) Triangle ABC is an equilateral triangle whose sides are of length 1 cm. If we select 5 points inside the triangle, prove that at least 2 of these points are such that the distance between them is less than $\frac{1}{2}$ cm.
- (iv) Prove that if 30 dictionaries in a library contains a total of 61,327 pages, then at least one of the dictionary must have at least 2045 pages.
- (v) Prove that every set of 37 positive integers contains at least two integers that leave the same remainder upon division by 36.
- (vi) What should be the minimum number of students so that at least two students have their last names begin with the same English letter.
- (vii) If five colors are used to paint 26 doors, show that at least six doors will have the same color.
- (viii) Show that if any 6 numbers from 1 to 10 are chosen, then two of them have their sum equal to 11.
- (ix) Show that if any $(n+1)$ numbers from 1 to $2n$ are chosen, then two of them will have their sum equal to $(2n+1)$.
- (x) Prove that 101 integers are selected from the set $S = \{1, 2, 3, \dots, 200\}$, then there are 2 integers such that one divides the other.
- (xi) Wilma operates a computer with a magnetic tape drive. One day she is given a tape that contains 5,00,000 words of four or fewer lower case letters. Can all words in the tape be distinct?
- (xii) A bag contains 12 pairs of socks(each pair of different colors). If a person draws the socks one by one at random, determine at most how many draws are required to get atleast one pair of matched socks.
- (xiii) Find the least number of ways of choosing 3 different numbers from 1 to 10 so that all choices have the sum.
- (xiv) Shirts numbered consecutively from 1 to 20 are worn by 20 students of a class. When any three of these students are chosen to be a debating team from the class, the sum of their shirt numbers is used as the code number of the team. Show that if any eight of the 20 are selected, then from these eight, we may form at least two different teams having the same code number.
- (xv) Show that every set of seven distinct integers includes two integers x & y such that atleast one of $x + y$ or $x - y$ is divisible by 10.